

30.1 Basic configuration

The DAC is configured using the following registers:

1. Power: The DAC is always connected to V_{DDA} . Register access is determined by PINSEL and PINMODE settings (see below).
2. Clock: In the PCLKSEL0 register ([Table 40](#)), select PCLK_DAC.
3. Pins: Enable the DAC pin through the PINSEL registers. Select pin mode for port pin with DAC through the PINMODE registers ([Section 8.5](#)). This must be done before accessing any DAC registers.
4. DMA: The DAC can be connected to the GPDMA controller (see [Section 30.4.2](#)). For GPDMA connections, see [Table 544](#).

30.2 Features

- 10-bit digital to analog converter
- Resistor string architecture
- Buffered output
- Selectable speed vs. power
- Maximum update rate of 1 MHz.

30.3 Pin description

[Table 538](#) gives a brief summary of each of DAC related pins.

Table 538. D/A Pin Description

Pin	Type	Description
AOUT	Output	Analog Output. After the selected settling time after the DACR is written with a new value, the voltage on this pin (with respect to V_{SSA}) is $VALUE \times ((V_{REFP} - V_{REFN})/1024) + V_{REFN}$. Remark: The DAC output is disabled when the device is in deep-sleep, power-down, or deep power-down mode.
V_{REFP} , V_{REFN}	Reference	Voltage References. These pins provide a voltage reference level for the ADC and DAC. Note: V_{REFP} should be tied to VDD(3V3) and V_{REFN} should be tied to V_{SS} if the ADC and DAC are not used.
V_{DDA} , V_{SSA}	Power	Analog Power and Ground. These should typically be the same voltages as V_{DD} and V_{SS} , but should be isolated to minimize noise and error. Note: V_{DDA} should be tied to VDD(3V3) and V_{SSA} should be tied to V_{SS} if the ADC and DAC are not used.

30.4 Register description

The DAC registers are shown in [Table 539](#). Note that the DAC does not have a control bit in the PCONP register. To enable the DAC, its output must be selected to appear on the related pin, P0.26, by configuring the PINSEL1 register. See [Section 8.5.2 “Pin Function Select Register 1 \(PINSEL1 - 0x4002 C004\)”](#). the DAC must be enabled in this manner prior to accessing any DAC registers.

Table 539. DAC registers

Name	Description	Access	Reset value ^[1]	Address
DACR	D/A Converter Register. This register contains the digital value to be converted to analog and a power control bit.	R/W	0	0x4008 C000
DACCTRL	DAC Control register. This register controls DMA and timer operation.	R/W	0	0x4008 C004
DACCNTVAL	DAC Counter Value register. This register contains the reload value for the DAC DMA/Interrupt timer.	R/W	0	0x4008 C008

[1] Reset value reflects the data stored in used bits only. It does not include reserved bits content.

30.4.1 D/A Converter Register (DACR - 0x4008 C000)

This read/write register includes the digital value to be converted to analog, and a bit that trades off performance vs. power. Bits 5:0 are reserved for future, higher-resolution D/A converters.

Table 540: D/A Converter Register (DACR - address 0x4008 C000) bit description

Bit	Symbol	Value	Description	Reset Value
5:0	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
15:6	VALUE		After the selected settling time after this field is written with a new VALUE, the voltage on the AOUT pin (with respect to V _{SSA}) is $VALUE \times ((V_{REFP} - V_{REFN})/1024) + V_{REFN}$.	0
16	BIAS ^[1]	0	The settling time of the DAC is 1 μs max, and the maximum current is 700 μA. This allows a maximum update rate of 1 MHz.	0
		1	The settling time of the DAC is 2.5 μs and the maximum current is 350 μA. This allows a maximum update rate of 400 kHz.	
31:17	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

[1] The settling times noted in the description of the BIAS bit are valid for a capacitance load on the AOUT pin not exceeding 100 pF. A load impedance value greater than that value will cause settling time longer than the specified time. One or more graph(s) of load impedance vs. settling time will be included in the final data sheet.

30.4.2 D/A Converter Control register (DACCTRL - 0x4008 C004)

This read/write register enables the DMA operation and controls the DMA timer.

Table 541. D/A Control register (DACCTRL - address 0x4008 C004) bit description

Bit	Symbol	Value	Description	Reset Value
0	INT_DMA_REQ	0	This bit is cleared on any write to the DACR register.	0
		1	This bit is set by hardware when the timer times out.	
1	DBLBUF_ENA	0	DACR double-buffering is disabled.	0
		1	When this bit and the CNT_ENA bit are both set, the double-buffering feature in the DACR register will be enabled. Writes to the DACR register are written to a pre-buffer and then transferred to the DACR on the next time-out of the counter.	
2	CNT_ENA	0	Time-out counter operation is disabled.	0
		1	Time-out counter operation is enabled.	
3	DMA_ENA	0	DMA access is disabled.	0
		1	DMA Burst Request Input 7 is enabled for the DAC (see Table 544).	
31:4	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

30.4.3 D/A Converter Counter Value register (DACCNTVAL - 0x4008 C008)

This read/write register contains the reload value for the Interrupt/DMA counter.

Table 542: D/A Converter register (DACR - address 0x4008 C008) bit description

Bit	Symbol	Description	Reset Value
15:0	VALUE	16-bit reload value for the DAC interrupt/DMA timer.	0

30.5 Operation

30.5.1 DMA counter

When the counter enable bit CNT_ENA in DACCTRL is set, a 16-bit counter will begin counting down, at the rate selected by PCLK_DAC (see [Table 40](#)), from the value programmed into the DACCNTVAL register. The counter is decremented Each time the counter reaches zero, the counter will be reloaded by the value of DACCNTVAL and the DMA request bit INT_DMA_REQ will be set in hardware.

Note that the contents of the DACCTRL and DACCNTVAL registers are read and write accessible, but the timer itself is not accessible for either read or write.

If the DMA_ENA bit is set in the DACCTRL register, the DAC DMA request will be routed to the GPDMA. When the DMA_ENA bit is cleared, the default state after a reset, DAC DMA requests are blocked.

30.5.2 Double buffering

Double-buffering is enabled only if both, the CNT_ENA and the DBLBUF_ENA bits are set in DACCTRL. In this case, any write to the DACR register will only load the pre-buffer, which shares its register address with the DACR register. The DACR itself will be loaded from the pre-buffer whenever the counter reaches zero and the DMA request is set. At the same time the counter is reloaded with the COUNTVAL register value.

Reading the DACR register will only return the contents of the DACR register itself, not the contents of the pre-buffer register.

If either the CNT_ENA or the DBLBUF_ENA bits are 0, any writes to the DACR address will go directly to the DACR register.

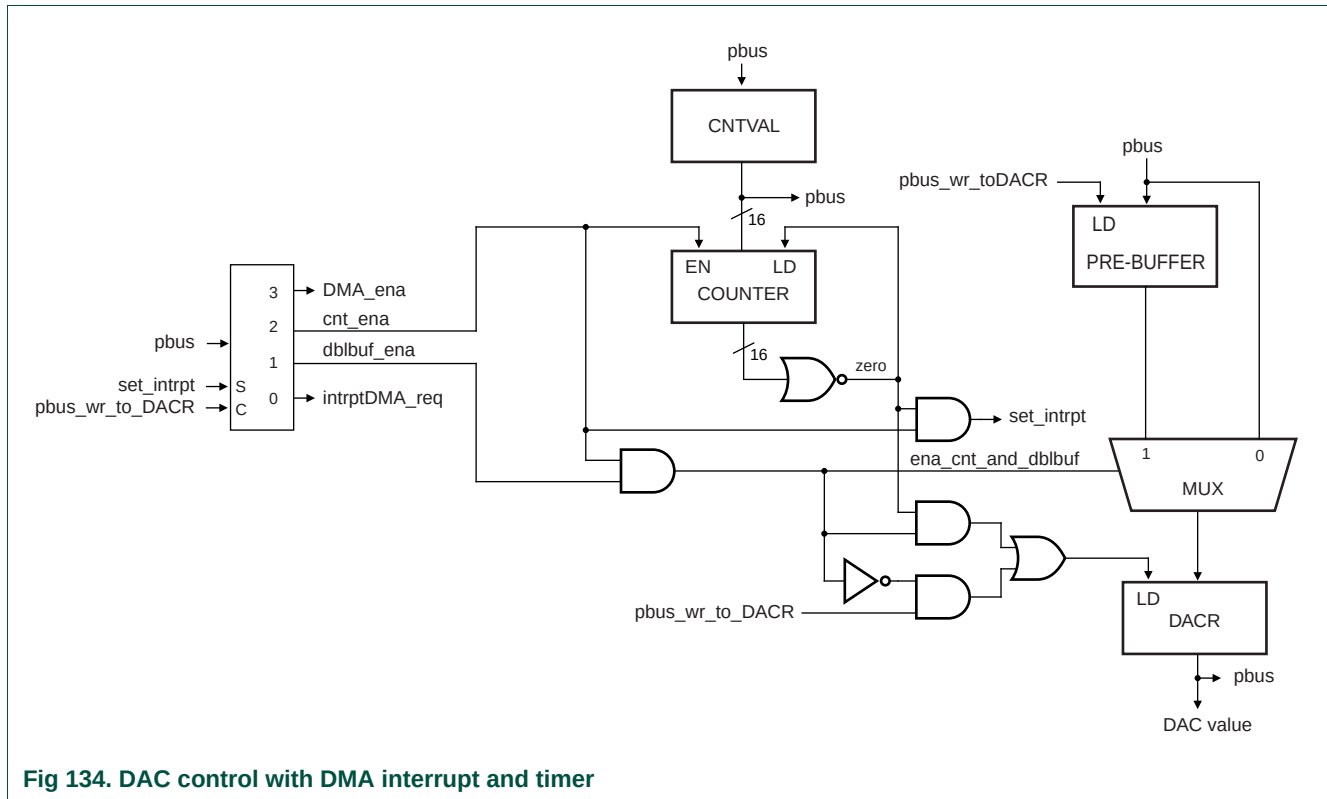


Fig 134. DAC control with DMA interrupt and timer